How Misinformation Diffuses on Online Social Networks: Radical Opinions, Adaptive Relationship, and Algorithmic Intervention

Mengyi Zhang, Qingxing Dong, and Xiaozhen Wu

***Abstract*—The advent of information distribution mechanism constituted by self-exploration, network neighbors, and especially algorithms, has aroused widespread concerns about the reinforcement of misinformation beliefs and the resulting polarization. However, few existing researches fully consider the inherent characteristics of misinformation (e.g. evoking repulsive effects), as well as the adaptive nature of social relationship or come to see the impacts of algorithmic interventions on online misinformation and the formation process of social groups. To comprehensively investigate the coevolution process of user misinformation beliefs and social relationships under algorithmic interventions, we proposed a novel model with configurations as follows: 1) a nonlinear social influence function constructed to reflect the process of reinforcing misinformation beliefs; 2) probabilities for the rewiring of links among individuals determined by their opinion distance and social distance; and 3) multiple algorithmic mechanisms reformulated, regarding five recommendation processes and the information distribution rules integrating three information sources. Such extensive numerical simulation experiments have revealed diversification, radicalization, and polarization of misinformation. We observe that the introduction of moderate repulsive interactions fosters the emergence of diverse opinions. In absence of algorithmic interventions, misinformation naturally evolves into radicalization, while the introduction of algorithmic interventions exacerbates polarization, particularly with extensive reliance on content-based recommendations and excessive allowance of distributed opinions from recommendations. It is noteworthy that we discover that encouraging recommendation based on predetermined information effectively reverses the trend of misinformation evolution. Our research contributes to clarifying the interaction between human behavior and artificial intelligence, as well as providing insights for misinformation supervision and governance.**

***Index Terms*—Complex network, misinformation diffusion, opinion dynamics, polarization, recommendation algorithm.**

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I. Introduction

On online social platforms, users can achieve information intake orders of magnitude beyond previous levels through three primary channels: self-exploration, virtual neighbors, and algorithms, with recommendation mechanisms being particularly prominent. This complex and diverse information distribution mechanism also provides an opportunity for misinformation to reach a wider audience, posing significant threats to society. Major concerns have been raised about the potential of algorithms in reinforcing the individual's conviction in misinformation [1], intensifying polarization [2], and producing cocoons [3]. Nevertheless, existing research often overlooks the inherent specificity of misinformation, such as the exposure to opposing opinions that creates potential risks of further solidifying ones' standpoint [4], as evidenced by the widely discussed "backfire effect" [5], [6] in sociological research. More importantly, this perspective biased towards the functionality of algorithms in filtering user-accessible content within opinion-sharing mechanisms lacks consideration of certain fundamental societal processes, such as homophily [7], structural constraints [8], and social influence [9].

There are even some inconsistent or contradictory empirical or causal evidence [10], [11], [12] hinting a largely unexplored landscape of the impact of algorithms on the diffusion of misinformation. This suggests the need for deeper investigation into the fundamental mechanisms and dynamic complexities of human–AI interaction systems. Simulation experiments offer a promising pathway in this regard, as well as allowing researchers to intuitively uncover unexpected scenarios of consensus or polarization in misinformation dynamics. Another problem is that the algorithmic mechanisms on today's platforms are notorious "black boxes" with billions of parameters, while the detailed algorithmic information and data remain inaccessible, making it challenging to conduct purely empirical research on these systems. If we can break down algorithms in a more granular way and simulate their effects, it not only helps us better understand these phenomena but also provides valuable insights to guide empirical research.

In this paper, we propose an agent-based dynamic model that captures three key ingredients: 1) *Reinforcement of misinformation beliefs*: a nonlinear function is constructed to reflect the social influence process, where controversial

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misinformation is designed to reinforce beliefs and induce more radical opinions. Based on this, we further introduce repulsive interactions (mechanism) to reproduce user's negative responds towards contrary opinions, namely the backfire effect. 2) *Adaptive nature of social relationship*: as dynamic processes that adapt to changing individual needs and media environments, the link relationships among individuals can be rewired based on opinion distance or social distance. 3) *Algorithmic information distribution mechanism*: we consider three sources of information distribution on platforms: exploration, neighbors, and recommendations, which mirrors reality more closely by acknowledging the persistent impact of individual decision-making (while also representing a pattern advocated in current and future scenarios). Additionally, we refine the recommendation mechanisms into a category of five types: content-based, users-based, relationship-based, popularity-based (as observed in bulletin boards), and predetermined information-based (i.e., the wide-discussed nudge mechanism [13]), so as to cover the underlying mechanisms of contemporary recommendation systems comprehensively.

Our contributions can be summarized as follows:

1) We establish a flexible opinion dynamic model of misinformation under algorithmic intervention. By incorporating the unique attributes of misinformation alongside natural social processes governing relationship evolution, and dissecting algorithmic information distribution mechanisms and recommendation mechanisms, we contribute to elucidating the interplay between human behavior and artificial intelligence.
2) Through extensive simulations conducted at both micro and macrolevels, we observe stable and regular evolutionary trends under varying combinations of factors. These outcomes reproduce scenarios of misinformation radicalization, polarization, and diverse distributions.
3) By fine-tuning the direction and magnitude of algorithmic intervention, we unveil effective intervention strategies for reversing evolutionary trends and combating the reinforcement of misinformation beliefs.
4) Utilizing a real-world network sourced from Reddit, our exploration reveals that the evolutionary trends and patterns observed closely parallel those identified in synthetic networks. This alignment provides robust support for both the microlevel observations and macrolevel assumptions of our model.

The remainder of this article is organized as follows: In Section II, related work is discussed and compared with our approach; in Section III, our proposed dynamic model and related algorithms are described; in Section IV, our experimental setups and obtained results are elaborated; and finally, in Section V, conclusions are drawn to outline suggestions for future work stemming from this study.

II. Related Research

A. Misinformation Diffusion Pattern

Early scholars examined the diffusion patterns of misinformation from a macroperspective, analyzing quantitative shifts among different groups at a collective level, and representatively introducing classical epidemic models and their various variants (such as SIR [14] and SIRS [15]). However, as information diffusion is fundamentally a process of individual opinion integration, there is a growing trend towards adopting a microperspective to introduce opinion dynamics models [16] that focus on how individuals in complex systems change their opinions in response to information received from the social environment. This approach facilitates the reproduction of the entire opinion evolution process under the influence of multifaceted variables and helps identify key factors influencing the opinion dynamic of misinformation, thereby formulating reasonable strategies to combat it.

Traditional frameworks for modeling opinions in continuous space commonly involve describing opinions' similarity in opinion space through bounded trust [17], [18] or homogeneity [19] based on agents' cultural features. However, these foundational models have two major drawbacks. On the one hand, these models assume that interactions between agents are solely based on individual opinions. The relative agreement model proposed by Deffuant attempts to overcome this limitation by focusing on the impact of random interactions [20]. Additionally, random contacts outside the opinion neighborhood are introduced as shown in literature [21], to mitigate like-minded interactions of bounded confidence models. On the other hand, they focus on the consensus induced by assimilative effects (i.e., attractive interactions), thus neglecting situations where distinct opinions held by two agents in cultural or religious conflicts are hardly reconciled [4]. Some scholars have gradually revised bounded confidence models by setting rejection thresholds [22] or considering different psychological profiles of individuals [23]. For misinformation, the focus of our study, existing research has shown that social psychological mechanisms, such as the worldview and familiarity backfire effects [24], can engender significant obstacles to assimilation or reconciliation among agents. Consequently, misinformation stands out as a quintessential example of information that requires attention to repulsive interactions. However, it has not yet received adequate focus within opinion modeling frameworks.

B. Algorithmic Information Distribution Mechanism

As for the impact of algorithms on the opinion dynamics of misinformation, research is still at an initial stage and remains on the periphery of the literature on opinion dynamics [25], with particularly scarce studies in the field of social sciences [26]. Peralta et al. directly consider algorithmic bias as a parameter embedded within a binary state opinion dynamics model, observing a range of rich phenomenology, including continuous and discontinuous phase transitions between stationary states such as consensus, coexistence, and polarization [27]. Related work has also assigned an external field to traditional voter models, representing user interactions with past histories to mimic content recommendations [28]. In another research, the effect of the algorithm is modeled by filtering interactions between individuals and their neighbors [29]. Pansanella et al., for example, simulated the homophilic rewiring process, observing

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the coevolution of opinions and network structures within an adaptive topology [30]. Additionally, some scholars have suggested increasing diversity within recommendation systems to ensure higher user engagement and alleviate filter bubbles [31]. Piao et al. considered the self-exploration mechanism in the real world, finding that it successfully pushes the system from an information cocoons toward diversification [3]. While the above studies have made significant attempts in simulating algorithmic mechanisms, they reflect varying definitions of what constitutes an "algorithmic mechanism." The terms "recommendation systems," "recommendation algorithms," "filtering," and "distribution" are often conflated, highlighting a broader challenge in accurately capturing the nuances of how these algorithms function in reality.

By far, we have reached a prevailing consensus that algorithms arrange information communication space around users and provide them with a subset of content recommendations (and potential connections) thereby deemed most pertinent. This essentially functions as an algorithmic information distribution mechanism, enabling clearer insights to be gleaned by tracing the evolution of information acquisition patterns. In offline societies, information acquisition primarily relies on interactions with geographical neighbors and occasional self-exploration; later it is expanded with the advent of news media. In the contemporary era of social media, similar to traditional media where gatekeeping starts with journalists [32], algorithmic gatekeeping already starts at source selection [33], while virtual social networks provide far more avenues for obtaining information from neighbors (i.e., friends we follow), along with convenient access to explore diverse information that is notably largely coming from personalized recommendation algorithms. Therefore, constituted of neighbors, exploration and recommendations, today's algorithmic information distribution mechanisms determine all the information we can receive. It's crucial to draw a strict distinction between "distribution" and "filtering" to avoid confusion: although their mechanisms may appear similar, "filtering" is more often used to describe the process of information recall in recommendation systems, which is just one part of information distribution mechanisms.

C. Recommendation Mechanisms

As the vital component of information distribution, recommended information is a culmination of various principles inherent in recommendation algorithms, but also a major source of bias. In traditional media, it is the editor's personal judgments and other personal factors that play a role in selecting news for newspapers, which may be highly subjective, based on the gatekeeper's own experience, attitudes, and expectations, leading to selection bias [34]. Similarly, in social media, recommendation algorithms attempt to collect personalized data on users' preferences and their past information consumption patterns, thus selectively displaying to them popular or similar materials. In practice, the primary objective of algorithm design in modern recommendation systems is to maximize user engagement and profit [35]. As often kept in business secrecy, the commercial recommendation systems' precise input data remains little known. Some studies have employed classic ensemble algorithms or link recommendation algorithms to compare the effects of different algorithms on opinion dynamics. Commonly used models include PageRank [36], ALS [37], implicit matrix factorization [38], Node2Vec [39], and random strategies [40]. However, these simplified ensemble algorithms differ significantly from the complex recommendation algorithms used in practice, resulting in poor transferability and minimal real-world applicability. Only by thoroughly dissecting the underlying mechanisms of recommendation platforms and breaking down the foundational recommendation rules can we effectively simulate real-world conditions and obtain persuasive observational results.

Insights gleaned from case studies and related leaks can help us uncover clues about the fundamental recommendation strategies used by various platforms. For example, Instagram's recommendation system primarily considers similar content, similar users, sender–receiver interaction relationships, and postspecific information (such as popularity) [41]. Major companies like Amazon have long adhered to collaborative filtering paradigms, determining user or item similarity based on past behaviors [42]. On Facebook, users are informed about why they see specific content, with reasons such as similar people, similar groups, similar content, or popular posts [43]. According to Twitter's open-source code from March 2023 [44], its recommendation sources include followed friends, user similarity, network similarity, and content interest similarity. There is also ongoing research that deconstructs the basic logic behind recommendation algorithms. For instance, Bellina et al. specifically discuss how user–user collaborative filtering algorithms influence agent opinions and behaviors [45]. Chavalarias et al. combine sentiment analysis with popularity evaluation tools to examine how emotional feedback and popularity algorithms affect social structures [25]. Pandey et al. explore how recommendation strategies based on user, opinion, and connection similarity impact creative processes and network cohesion [46].

One particular recommendation strategy, often obscured by commercial platforms, is recommending the same predetermined information to everyone. However, it is also the most direct way to impose platform bias, which may trigger a storm of misinformation and cause serious political manipulation consequences if not handled properly or even exploited for propaganda purposes [47]. This is closely related to the nudge theory [48] in behavioral economics and companies controlling media outlets or external players through paid advertisement or by the use of bots can easily introduce this type of nudge. Perra et al. simulated a situation where a small fraction of posts in a user's timeline on social media is manipulated to promote certain opinions, observing that public opinions can relatively quickly shift in the direction of the pushed opinion, ultimately completely altering the status quo [49].

Building on both real-world applications and previous research, we do not aim to recreate any specific ensemble algorithm in detail. Instead, we summarize and model five fundamental types of recommendation algorithms: content-based, user-based, relationship-based, popularity-based, and

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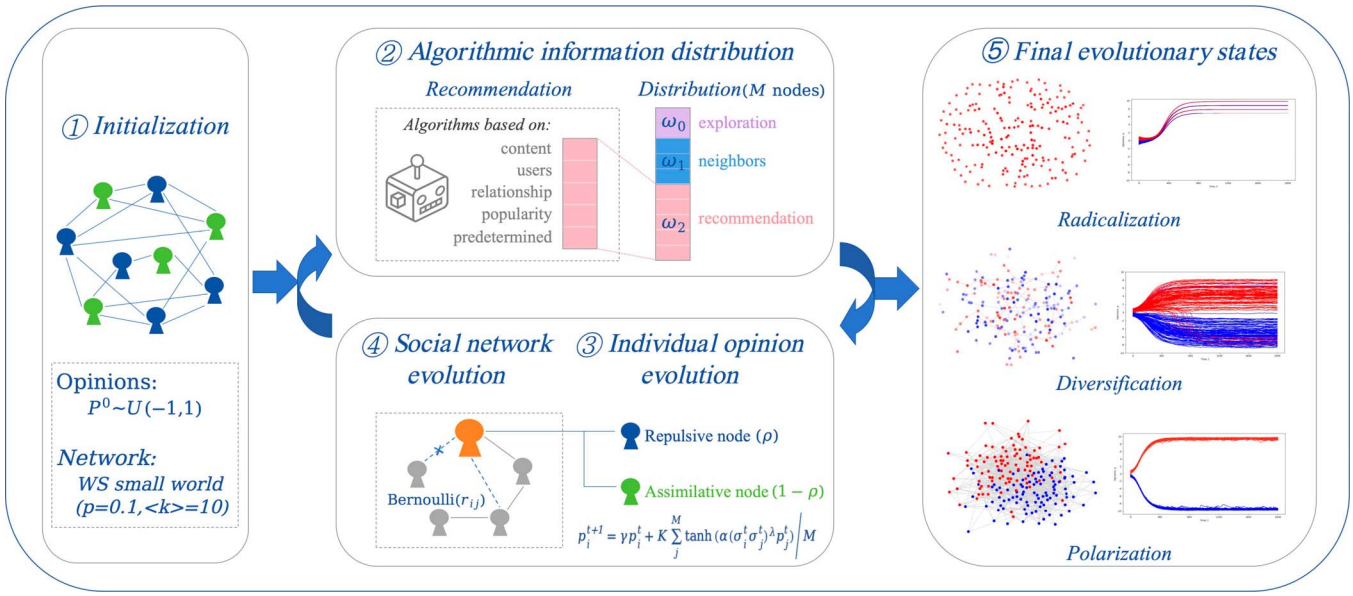

Fig. 1. Framework of modeling misinformation opinion dynamics under algorithmic intervention.

predetermined information-based. Furthermore, recognizing the current state of recommendation systems, which utilize multiple data sources to generate more robust inferences, we account for how various recommendation outputs are combined to generate the final output [50]. To the best of our knowledge, our proposed framework is the first to deconstruct the foundational mechanisms of algorithmic distribution and recommendation, while creating a hybrid scenario that incorporates three types of information distribution sources and five fundamental recommendation algorithms. This approach allows us to granularly reveal the influence of each component on the dynamics of misinformation opinions without oversimplifying any aspect.

III. Model

A. Overview of the Model

As shown in Fig. 1, our modeling framework consists of five stages:

1) *Initialization:* Within the population, nodes are categorized into two distinct types, each exhibiting varied responses to opinions. Every individual is attributed a type, along with an initial stance towards misinformation, initial social network relationships, and a set of prior opinions.
2) *Algorithmic information distribution:* A fixed proportion of neighbor nodes, exploration nodes, and recommended nodes are distributed to each node, which constitutes the interaction node set. Recommended nodes are determined based on different recommendation rules, sorted and filtered. This iterative process occurs at each time step.
3) *Individual opinion evolution:* Influenced by the attitudes towards misinformation in the interaction node set, each node updates its opinion in accordance with its inherent response type. This iterative process occurs at each time step.
4) *Social network evolution:* The rewiring between each node and those within the newly encountered node set has its probability calculated and is implemented according to Bernoulli probability. Upon establishment of a new connection, one existing connection of the node is randomly removed. This iterative process occurs at each time step.
5) *Final evolutionary states:* The evolution of opinions and relationships across all time steps is followed to ascertain the final evolutionary state of the population and quantify the levels of radicalization and polarization within the group.

B. Related Definitions

Consider a set $X = \{x_1, x_2, \ldots, x_n\}$ consisting of n individuals participating in the opinion evolution process, with $N = \{1, 2, \ldots, n\}$ denoting the index of elements in the set, and $t (t \in \mathbb{N})$ denoting the interaction round (time step) during the evolution process. Assume each individual has an attitude towards a certain item of misinformation, which can be represented by a one-dimensional opinion variable $p_i^t (p_i^t \in (-\infty, +\infty))$, indicating the opinion of x_i towards the misinformation at time t. Let P_N^t be the set of opinions of all individuals at time t. The opinions of individuals towards the misinformation can be positive or negative, with the sign of p_i^t (denoted by σ_i^t) indicating the individual's belief position on the misinformation (i.e., believing or not believing). Specifically, σ takes values of 1 or -1, which helps determine whether the stance of two interacting nodes are aligned (e.g., $\sigma_i^t \sigma_j^t = -1$ suggests that individuals x_i and x_j belong to opposing belief camps during interaction at time t). The absolute value of p_i^t, $|p_i^t|$, captures how radical an individual's opinion is. Additionally, each individual x_i has a set of prior opinions denoted by O_i. The social network for evolution is denoted by $G(X, E)$, where the edges between nodes are represented by E.

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Definition 1 (Type of Node Interaction): Each individual may exhibit different responses when confronted with contrary opinions, classified into two categories:

1) *Assimilative node:* When faced with opinions that differ significantly from their own, the individual tends to assimilate towards the direction of the opposing opinion to mitigate disagreement.
2) *Repulsive node:* When faced with opinions that differ significantly from their own, the individual instead tends to strengthen their own opinion, exacerbating the existing disagreement.

Definition 2 (Network Connectivity): Let $A^t = (a_{ij}^t) \in \{0,1\}_{n\times n}$ be the adjacency matrix representing the social network relationships among individuals at time t

$$a_{ij}^t = \begin{cases} 1, (x_i, x_j) \in E \\ 0, (x_i, x_j) \notin E. \end{cases} \tag{1}$$

If there is a connection between individuals x_i and x_j, $a_{ij}^t = 1$; otherwise, $a_{ij}^t = 0$. Specifically, A^0 represents the adjacency matrix of the initial social relationship network among individuals.

Definition 3 (Degree of Centrality): Individual degree of centrality at time can be defined as the number of its neighbors (links/friends), i.e.

$$\mathrm{CD}_i^t = \sum_{j=1}^{N} a_{ij}^t. \tag{2}$$

$\mathrm{CD}_i^t \in [0, n-1]$. The higher the degree of a node, the more important it is in the network.

Definition 4 (Social Distance): The social distance between individual x_i and other nodes at time t can be represented by their shortest paths in the social network, i.e.

$$\mathrm{SoDis}_{ij}^t = g_{\text{short_path}}(G, x_i, x_j) \tag{3}$$

where $g_{\text{short_path}}()$ represents the function for finding the shortest path.

Definition 5 (Opinion Distance): The opinion distance between individual x_i and other nodes at time t can be represented by the difference in their opinions towards the-misinformation, i.e.

$$\mathrm{OpDis}_{ij}^t = \left|p_i^t - p_j^t\right|. \tag{4}$$

Definition 6 (Neighbor Set): At time t, the neighbor set of individual x_i is defined as

$$\mathrm{Nei}_{x_i}^t = \{x_j \mid a_{ij}^t = 1\}, \forall i, j \in N. \tag{5}$$

Therefore, $\#\mathrm{Nei}_{x_i}^t (\#\mathrm{Nei}_{x_i}^t \in [0, n-1])$ represents the number of neighbors of the individual x_i.

Definition 7 (Common Neighbors): Based on the neighbor set of an individual, the common neighbors between individual x_i and x_j at time t are defined as

$$\mathrm{CN}_{ij}^t = \{\mathrm{Nei}_{x_i}^t \cap \mathrm{Nei}_{x_j}^t\}, \forall i, j \in N. \tag{6}$$

Therefore, $\#\mathrm{CN}_{ij}^t$ represents the number of common neighbors between individuals x_i and x_j.

Definition 8 (Recommendation Set): At time t, the set of potential recommended nodes designed for individual x_i is denoted as

$$\mathrm{Rec}_{x_i}^t = \{x_k \mid f_{\text{rec_}}(x_i, t) \cap x_k \notin \mathrm{Nei}_{x_i}^t\}, \forall i, k \in N \tag{7}$$

where $f_{\text{rec_}}(x_i, t)$ represents the function implementation of the recommendation algorithm and return the results sorted in descending order according to specific recommendation rules. This article involves five recommendation algorithms, namely $f_{\text{rec_content}}(x_i, t)$, $f_{\text{rec_users}}(x_i, t)$, $f_{\text{rec_relationship}}(x_i, t)$, $f_{\text{rec_popularity}}(x_i, t)$, and $f_{\text{rec_predetermined}}(x_i, t)$. $\mathrm{Rec}_{x_i}^t$ does not include neighbor nodes.

Definition 9 (Opinion Popularity): The mainstream opinion of the social group at a certain time t reflects the popularity of opinions, as represented by the average attitude tendency of the entire group at that time, i.e.

$$\overline{P_N^t} = \frac{1}{N} \sum_{n=1}^{N} p_n^t. \tag{8}$$

C. Algorithmic Information Distribution

In the evolutionary system, individuals are required to interact with other nodes and adapt their opinions and social network relationships accordingly. However, as the size of the evolving network expands, individuals encounter such challenges in attaining a comprehensive awareness of the opinions held by all members within the group that they can only access and process a finite amount of information owing to cognitive constraints and attention budget [51]. To address this, platforms regulate the number of information feeds displayed on a user's screen, providing a fresh batch of information with each refresh. Similarly, our model assumes that at any given time, any individual can interact with M nodes (represented by the set $I_{\text{distributed}}$), including m_0 exploration nodes, m_1 neighbor nodes, and m_2 recommendation nodes. This is precisely the distribution rule followed by the platform, where each type of node is recalled to specific users according to fixed proportions ω_0, ω_1, and ω_2

$$M = \omega_0 m_0 + \omega_1 m_1 + \omega_2 m_2 \tag{9}$$

where $\omega_0 + \omega_1 + \omega_2 = 1$.

According to *Definition 6*, at time t, an individual has a total of $\#\mathrm{Nei}_{x_i}^t$ neighbors. The neighbor nodes used for interaction are randomly selected from $\mathrm{Nei}_{x_i}^t$, which consists of m_1 $(0 \le m_1 \le \#\mathrm{Nei}_{x_i}^t)$ nodes, denoted by set I_{nei}. It is worth noting that if at time t the value of m_1 is greater than $\#\mathrm{Nei}_{x_i}^t$, the actual neighbors interacted with individual x_i will be limited to $\#\mathrm{Nei}_{x_i}^t$, i.e. $m_1 = \#\mathrm{Nei}_{x_i}^t$.

Additionally, individual x_i can interact with recommendation nodes from specific recommendation algorithms. This study summarized five fundamental recommendation rules, with the results of each algorithm being presented in descending order, as shown in Table I specifically:

1) $f_{\text{rec_content}}(x_i, t)$*:* The content-based recommendation algorithm calculates the similarity of opinion between individuals at time t;

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TABLE I
LIST OF RECOMMENDATION ALGORITHMS

Denote	Formula
$f_{\text{rec_content}}$	$\text{Rec}_{x_i}^t = \text{sort}\left(\left\|p_i^t - p_j^t\right\| \text{ for } j \text{ in } X_{-i}\right)$
$f_{\text{rec_users}}$	$\text{Rec}_{x_i}^t = \text{sort}\left(\left\|o_i^t - o_j^t\right\| \text{ for } j \text{ in } X_{-i}\right)$
$f_{\text{rec_relationship}}$	$\text{Rec}_{x_i}^t = \text{sort}\left(\frac{\#\text{CN}_{ij}^t}{\max\left(\text{CD}_i^t, \text{CD}_j^t\right)} \text{ for } j \text{ in } X_{-i}\right)$
$f_{\text{rec_popularity}}$	$\text{Rec}_{x_i}^t = \text{sort}\left(\left\|p_j^t - \overline{P_N^t}\right\| \text{ for } j \text{ in } X_{-i}\right)$
$f_{\text{rec_predetermined}}$	$\text{Rec}_{x_i}^t = \text{sort}\left(\left\|p_j^t - o'\right\| \text{ for } j \text{ in } X_{-i}\right)$

2) $f_{\text{rec_users}}(x_i, t)$: The users-based recommendation-algorithm recommends nodes whose prior behavior (opinions) is similar, with the prior opinion set $O_{n\times n}$ simulating user prior behavior and remaining constant over time in this study;
3) $f_{\text{rec_relationship}}(x_i, t)$: The relationship-based recommendation algorithm recommends nodes with the most similar network structure, meaning if individual x_i and x_j share a common friend x_k, then x_i and x_j are likely to become friends (establish a connection) based on the classic triadic closure principle [52], so the more common neighbors two nodes have, the more similar they are and thus more suitable for recommendation. In this study, the similarity between x_i and x_j in terms of network structure at time is calculated using the high degree of interconnectivity (HDI) [53] index as a metric for node similarity—specifically, when $\max(\text{CD}_i^t, \text{CD}_j^t)$, it implies that both individuals x_i and x_j have no edges established with other individuals, and their similarity is assumed to be 0;
4) $f_{\text{rec_popularity}}(x_i, t)$: The popularity-based recommendation algorithm recommends nodes whose opinions are most similar to the most popular opinion at time t;
5) $f_{\text{rec_predetermined}}(x_i, t)$: The predetermined-based recommendation algorithm calculates the similarity between individual opinions and the predetermined opinions set by the platform, the latter denoted by o' and constant over time.

According to *Definition 8*, the m_2 recommendation nodes that finally reach the user are the top m_2 nodes sorted by specific recommendation algorithms, denoted by set I_{rec}. So if a hybrid recommendation scenario is to be simulated, the proportion ξ for each recommendation algorithm can be set within the returned m_2 nodes.

Finally, in addition to neighbor nodes and recommendation nodes, m_0 exploration nodes are randomly selected from the remaining nodes (denoted by set I_{explore}) and distributed to each user.

D. Individual Opinion Evolution

Inspired by previous research on social influence [54], [55] and dynamics of radicalization [1], [56], we assume that the evolution of an individual's opinion on misinformation follows the following rules:

$$p_i^{t+1} = \gamma p_i^t + K \sum_{j=1}^{M} \tanh(\alpha(\sigma_i^t \sigma_j^t)^\lambda p_j^t)/M. \tag{10}$$

γ is a decay factor, indicating that in the absence of social interactions, an individual's opinion on misinformation tends to attenuate towards a neutral state. Parameter K controls the strength of influence of M interactive nodes on the opinion of focal agent x_i. The hyperbolic tangent (tanh) introduces nonlinearity to the social influence of opinions and imposes a bound on the degree of influence individuals have on each other. Parameter α ($\alpha > 0$) governs the shape of this influence function. Higher values of α are associated with more controversial topics, where individuals pay more attention to the topic and thus are more susceptible to social influence [1]. It is worth noting that our evolution rules can also simulate consistently stubborn individuals by manipulating γ and K, ensuring that p_i^{t+1} remains equal to p_i^t. However, in real-world controversial scenarios involving misinformation, such stubborn individuals are rare. While people may strongly adhere to their beliefs or affiliations, these beliefs are unlikely to remain entirely unchanged; there is typically some fluctuation over time, either reinforcing or weakening. Therefore, we do not include stubborn individuals in our study.

Here, we further derive the upper and lower bounds of p_i. First, the difference in opinion between two adjacent time steps for x_i is given by $\triangle p_i = p_i^{t+1} - p_i^t = (\gamma - 1)p_i^t + K\sum_{j=1}^{M} \tanh(\alpha(\sigma_i^t\sigma_j^t)^\lambda p_j^t)/M$. As this difference approaches zero, we obtain the extreme values of p_i. In this model, we assume that x_i's opinion is updated based on the average influence of M interacting nodes in an unweighted network, thus $K\sum_{j=1}^{M} \tanh(\alpha(\sigma_i^t\sigma_j^t)^\lambda p_j^t)/M$ is constrained to the interval $[-K, K]$. Therefore, $\max|\triangle p_i| = |p_i^t(\gamma - 1) + K|$. When p_i^t approaches $|K/(1-\gamma)|$, we obtain the maximum and minimum values of p_i. Hence, γ and K effectively limit the range of x_i's opinion, i.e., $p_i < |K/(1-\gamma)|$. Referring to the parameter settings in the literature [56], we set $\gamma = 0.99$ and $K = 0.1$ in all experiments in Section IV, which restricts p_i to the interval $[-10, 10]$.

We also consider the possibility of both assimilative and repulsive effects simultaneously in the model. We use the parameter ρ to control the proportion of repulsive nodes in the population. When the signs of the opinions of two nodes are the same, it indicates that they are at least in the same camp and there is no significant disagreement. However, when $\sigma_i^t\sigma_j^t = -1$, it indicates that the two interacting nodes are in opposing states, reflecting significant disagreement. On the one hand, if the focal node is an assimilative node, then setting $\lambda = 0$ allows the node to move in the opposite opinion direction, reducing the opinion divergence; on the other hand, if the node is repulsive, then setting $\lambda = 1$ reproduces the backfire effect of misinformation, causing the node to strengthen its own opinion, leading to further divergence in opinions.

We present a schematic in Fig. 2. Suppose agent x_i has an opinion value of 5.0 at time step t, and it interacts with a total of ten nodes: two exploration nodes, three neighboring nodes and

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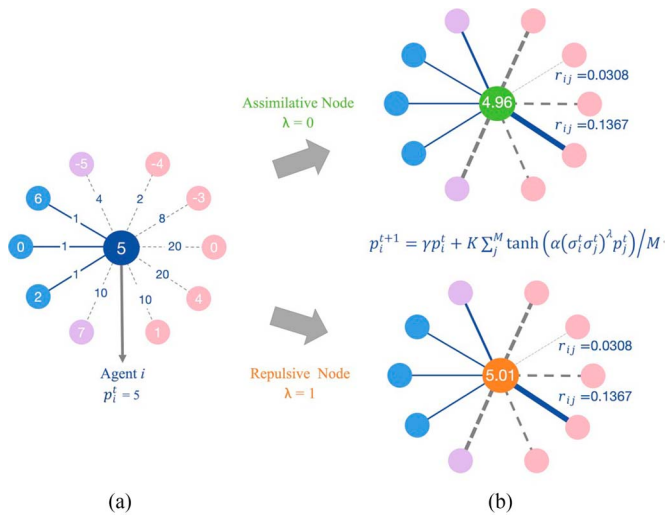

Fig. 2. Schematic of opinion and network evolution. The solid circles represent interacting nodes, with the focal node in dark blue, while blue, purple, and pink circles represent nodes from neighbors, exploration, and recommendations, respectively. The numbers inside the nodes indicate the opinion values. Solid blue lines between nodes represent direct connections, while gray dashed lines indicate unconnected nodes. The values in the middle of the lines represent the shortest path length, and the thickness of the lines denotes the probability of reconnection. Subplot (a) shows the opinion of focal node x_i at time t and its interaction with ten other nodes and (b) displays the updated opinion value of the focal node and the updated links at time $t+1$.

five recommendation nodes. Depending on the type of node, the opinion value of x_i at time step $t+1$ is updated according to slightly different rules. The final updated opinion value, influenced by the "social influence" of these ten interacting nodes, is 4.96 and 5.01, respectively.

E. Social Network Evolution

Considering the dynamic nature of real-world social networks, dynamic social networks with rewiring mechanisms have been widely studied by scholars, with previous models typically discussing the relationship evolution rules based on random rewiring [37], [40], opinion distance in bounded confidence models [18], and homophily of opinions [1]. However, we argue that social connections between individuals are not solely based on random probabilities or consensus on a single viewpoint. Santos et al. proposed a method of calculating rewiring probabilities based on network structural similarity [56]. However, in real-world decision-making, individuals cannot access or compute such detailed structural information. Inspired by the primary constraints on human relationship formation in traditional offline societies, we propose that relationships among people are formed based on both opinion distance and social distance simultaneously. Therefore, for each individual x_i, the connection probability r_{ij} between it and each newly encountered node (i.e., from exploration or recommendation nodes) x_j can be computed as

$$r_{ij} = \mu \frac{|p_i - p_j|^{-\eta}}{\sum_j^M |p_i - p_j|^{-\eta}} + \phi \frac{l_{ij}^{-\eta}}{\sum_j^M l_{ij}^{-\eta}}. \tag{11}$$

In this equation, the opinion distance between individual x_i and x_j is represented by their opinion distance $|p_i - p_j|$, while the social distance is represented by the shortest path in the social network, i.e., l_{ij}. The parameter η controls the power-law decay of rewiring probability with respect to the two distance rules, while μ and ϕ ($\mu + \phi = 1$) respectively control the relative influence strength of the two distance rules.

As shown in Fig. 2, at time t, agent x_i interacts with new nodes that are not directly connected, aside from its neighbor nodes. The probability of rewiring with these newly encountered nodes at the next time step depends on their opinion values and the shortest path distance. We also display the maximum and minimum rewiring probabilities calculated for this case. Ultimately, whether a connection is established between individuals x_i and x_j depends on a Bernoulli trial with a probability of r_{ij}. It is important to note that at each time step, we maintain a constant number of edges for each node, meaning that when a new link is successfully created, another link between the node and one of its original neighbors is randomly deleted. By doing so, we eliminate the influence of network factors (such as size and density) on the opinion dynamics, ensuring that any observed results are solely attributable to model parameters or algorithmic interventions.

IV. Experiments and Results

A. Experimental Procedure and Parameter Setting

In this study, we conduct simulation experiments to demonstrate the dynamics of opinions and networks under different configurations. We consider a Watts–Strogatz small-world network with $N = 200$ nodes (rewiring probability $p = 0.1$, average degree $< k > = 10$), where each individual starts with moderate initial conditions (opinions uniformly sampled from the interval $[-1, 1]$). Additionally, each agent is randomly assigned a prior opinion value from a uniform distribution to represent a user attribute. During the evolution process, each agent interacts with up to ten nodes at each time step (i.e., $M = 10$), which approximates the average number of times users engage with posts during scrolling sessions on social blogging platforms [51]. We observe the changes in opinions and networks for at least 2000 time steps in each experiment. To reduce random errors, the results of each parameter experiment are averaged over at least 50 independent runs.

To analyze the nature of the evolution results, we select several observation metrics. First, interested in the levels of radicalization and polarization of attitudes towards misinformation, we use the standard deviation (SD) of opinions and the mean of absolute value of opinions similar to practices in the literature [1] and [56], so as to quantify the levels of radicalization and polarization in the final state. Unlike the traditional consensus decision-making fields where the time required to reach consensus is an important indicator of group decision-making effectiveness, in our simulation model the group dynamics do not necessarily converge to consensus as a steady state. Referring to automatic control theory, if the corresponding curve reaches and maintains within an allowable error range (usually 5%) of the stable state value, we consider the dynamic system to

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TABLE II
LIST OF PARAMETERS

Parameter	Property	Domain	Values Setting
Opinion evolution parameters			
γ	Opinion decay factor	$0 \leq \gamma \leq 1$	{0.99}
K	Social interaction strength	$K \in \mathbb{R}^+$	{0.1}
α	Opinion social influence/controversialness	$0 \leq \alpha \leq 1$	{0.1, 0.2, 0.3,…,0.9, 1}
M	Number of distributed (interaction) nodes	$M \in \mathbb{N}^+$	{10}
ρ	Proportion of repulsive nodes in the population	$0 \leq \rho \leq 1$	{0.1, 0.2, 0.3,…,0.9, 1}
Network evolution parameters			
η	Power-law decay strength of rewiring probability	$\eta \in \mathbb{R}^+$	{0, 0.2, 0.4,…,2.5, 3}
μ	Relative influence weight of rewiring based on opinion distance	$0 \leq \mu \leq 1$	{0.1, 0.2, 0.3,…,0.9, 1}
ϕ	Relative influence weight of rewiring based on social distance	$0 \leq \phi \leq 1$	{0.1, 0.2, 0.3,…,0.9, 1}
Algorithmic intervention parameters			
ω_0	Proportion of distributed nodes from exploration	$0 \leq \omega_0 \leq 1$	{0.1, 0.2, 0.3,…,0.9, 1}
ω_1	Proportion of distributed nodes from neighbors	$0 \leq \omega_1 \leq 1$	{0.1, 0.2, 0.3,…,0.9, 1}
ω_2	Proportion of distributed nodes from recommendations	$0 \leq \omega_2 \leq 1$	{0.1, 0.2, 0.3,…,0.9, 1}
ξ_1	Proportion of content-based recommendation nodes in hybrid recommendation scenario	$0 \leq \xi_1 \leq 1$	{0, 0.1, 0.2,…,0.9, 1}
ξ_2	Proportion of users-based recommendation nodes in hybrid recommendation scenario	$0 \leq \xi_2 \leq 1$	{0, 0.1, 0.2,…,0.9, 1}
ξ_3	Proportion of relationship-based recommendation nodes in hybrid recommendation scenario	$0 \leq \xi_3 \leq 1$	{0, 0.1, 0.2,…,0.9, 1}
ξ_4	Proportion of popularity-based recommendation nodes in hybrid recommendation scenario	$0 \leq \xi_4 \leq 1$	{0, 0.1, 0.2,…,0.9, 1}
ξ_5	Proportion of predetermined-information-based recommendation nodes in hybrid recommendation scenario	$0 \leq \xi_5 \leq 1$	{0, 0.1, 0.2,…,0.9, 1}
o'	Opinion value of predetermined information	$o' \in \mathbb{R}$	{0, -0.0001, 0.0005, -0.001, -0.005, -0.01,…,-8, -10}
Simulation parameter			
N	Population size	$N \in \mathbb{N}^+$	{200}
T	Maximum time step	$T \in \mathbb{N}^+$	{2000,3000}
R	Number of independent runs	$R \in \mathbb{N}^+$	{50}

have reached a stable state. Similarly, within the maximum time step range we set, if we observe a stable evolutionary trend, we calculate the second-order difference in radicalization levels with respect to the final stable state at any given time during the evolutionary process. This allows us to measure the minimum adjustment time t_s, which represents the fastest time to reach the final stable state. All parameters and details in the model are summarized in Table II.

All experiments can be divided into five parts. First, we reveal the entire process of individual opinion evolution under different basic scenarios from a microperspective (Section IV-B1), which facilitates the observation of the validity of the model. Second, in order to cover as many cases as possible and make the model applicable to different users and platform scenarios, we need to explore the effects of various parameters involved in the model and changes in different configurations on the performance indicators. Therefore, we focus on the impact of opinion evolution parameters (i.e., opinion social influence parameter α and the proportion of repulsive nodes ρ) in Sections IV-C1 and IV-C2. Third, we investigate the relative influence weight of link rewiring rules Section IV-D1, as well as the influence strength (Section IV-D2) to observe the effects of network evolution parameters. Fourth, we concentrate on the intervention effects of algorithms, mainly discussing the utility of each basic recommendation algorithm under hybrid recommendation scenarios (Section IV-E1), and the results of adjusting the algorithmic distribution mechanism (Section IV-E2), along with Section IV-E3 particular set up to explore the positive effects of leveraging recommendation algorithms based on predetermined opinions to combat misinformation. To reduce random errors, the results of each parameter experiment are averaged over at least 50 independent runs.

Finally, we conducted experiments on our model using a real-world social network dataset obtained from Reddit, aiming to establish a connection between our extensive ideal simulations and empirical data. Reddit, a prominent social media platform with substantial traffic that revolves around interest-based communities known as subreddits. Users freely engage in discussions by posting comments or replies within these subreddits, thereby providing us with a tangible network of opinion interactions. We obtained this dataset from [57], encompassing all submissions to the subreddit "r/news" from 2016 to 2020, and categorized the content into different topic categories through reliable topic clustering techniques [57]. To mitigate sparsity issues and reduce computational costs, we focused specifically on data related to the "TECH" topic discussions in the year 2020. Additionally, users were filtered based on their activity levels, excluding from the dataset with fewer than 25 submissions on r/news within that year, or those identified as belonging to known Reddit bot lists. Based on the final set of users obtained and all interaction records, we conceptualized users as nodes within the network; if user u replies to a message posted by user v, we define an edge (i.e., $a_{uv} = 1$), thereby constructing a network of opinion

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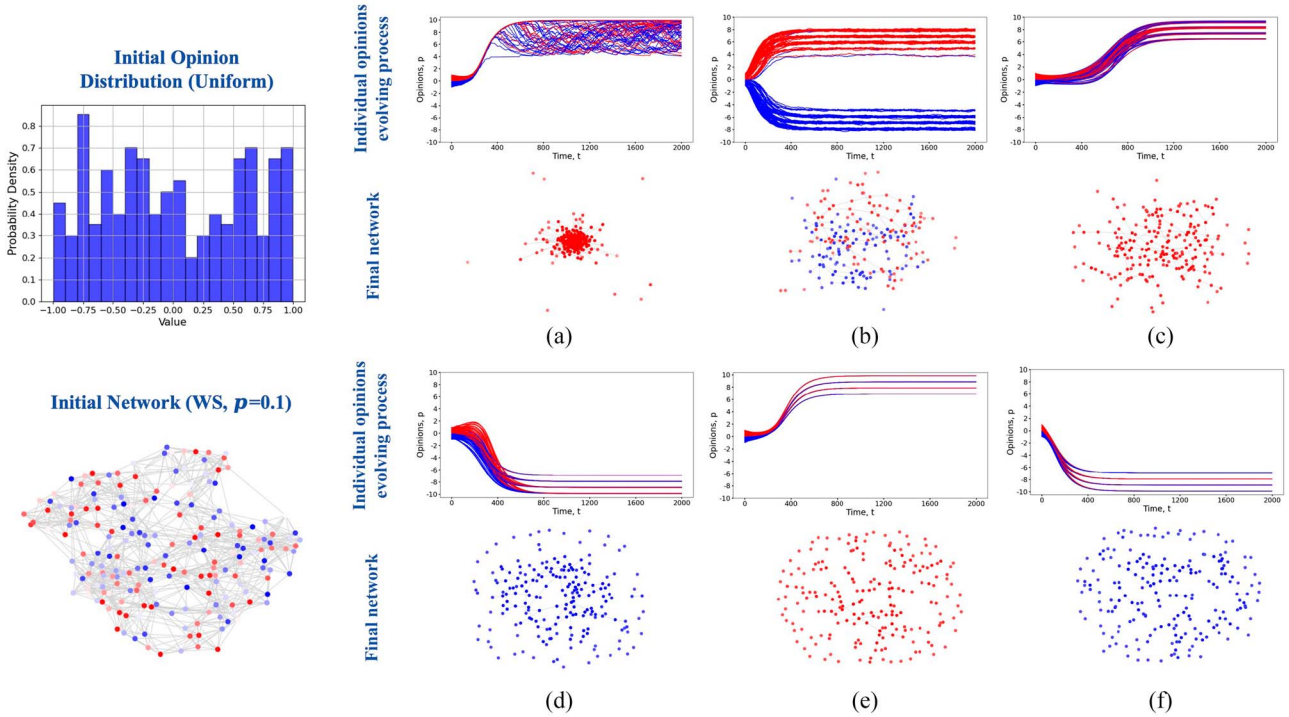

Fig. 3. Temporal evolution of the agents' opinions under different recommendation intervention strategies. The left side of the figure displays the initial conditions. The six subplots on the right (a)–(f) correspond to the temporal evolution over 2000 time steps under the intervention of recommendation algorithms based on content, users, relationships, popularity, and predetermined opinions (each subplot's upper half) as well as the final network structure (the lower halves). In the networks, red nodes represent positive opinions, blue nodes denote negative opinions, and white nodes indicate neutrality. The color intensity increases with the absolute value of opinion scores. Other parameters: $N = 200, K = 0.1, \gamma = 0.99, \rho = 1, \mu = 1, \eta = 1$.

interactions. Ultimately, this network comprises 4585 nodes and 6491 edges.

B. Model Validity

1) Entire Evolutionary Processes Under the Intervention of Different Recommendation Algorithms: In this experiment, evolving from the same initial condition, six basic systems are set up (Sections II–V are also based on them): no recommendation algorithmic intervention, recommendation intervention based solely on content, solely on users, solely on relationship, solely on popularity, and solely on predetermined opinion. In Fig. 3, we see three types of evolution patterns: continuous oscillation within a one-sided radicalization range [see Fig. 3(a)], one-sided radicalization [Fig. 3(c)–3(f)] or polarization (a polarized state characterized by a bimodal distribution of opinions emerges) [Fig. 3(b)]. The different evolutionary patterns preliminarily reflect the trend of influence of different recommendation algorithms. For instance, in Fig. 3(a), without intervention from recommendation mechanisms, users receive information only from neighbors and exploration nodes, ultimately leading to a state of continuous oscillation that tends towards radicalization but remaining unstable.

With the introduction of recommendation mechanisms, the evolutionary situation stabilizes, but the effects of different recommendation algorithms vary greatly. Specifically, content-based recommendation algorithms seem most likely to induce polarization [Fig. 3(b)]; user-based recommendation algorithms [Fig. 3(c)] and popularity-based recommendation algorithms [Fig. 3(e)] promote the stability of radicalization, but the former leads to fewer isolated modules. On the other hand, relationship-based [Fig. 3(d)] and predetermined opinion-based [Fig. 3(f)] recommendation algorithms have the power to disrupt the original evolutionary direction, shifting the group's overall belief from leaning towards believing to leaning towards disbelieving misinformation.

C. Impact of Opinion Evolution Parameters

1) Impact of Opinion Social Influence α on the Final Stable State: In the previous research, it was highlighted that the representative parameter α of opinion social influence can determine whether individuals reach a state of neutral consensus or radicalization [56], to which similar results are also observed in our experiments. If only extreme opinions generate social influence (i.e., low α), regardless of the presence of algorithmic intervention, the population tends towards a state of neutral consensus. However, if moderate social opinions also produce strong social influence (found in a grid search at approximately $\alpha > 0.2$), opinions become radicalized. Under the influence of different algorithmic interventions, three different stable evolutionary patterns observed in Section IV-B1 emerge. Additionally, when simulated under different initial conditions, within a certain range, larger values of α shorten the minimum adjustment time to reach the final stable state, which holds true for all intervention scenarios (as shown in Fig. 4). As mentioned

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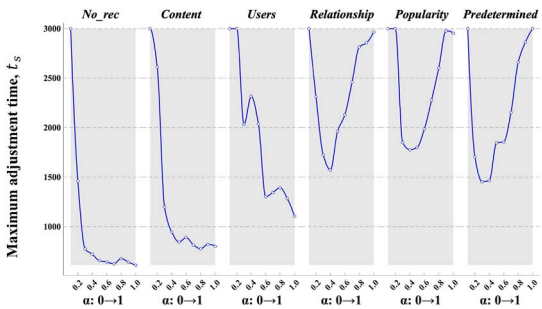

Fig. 4. Variation of maximum adjustment time t_s with α under six fundamental systems.

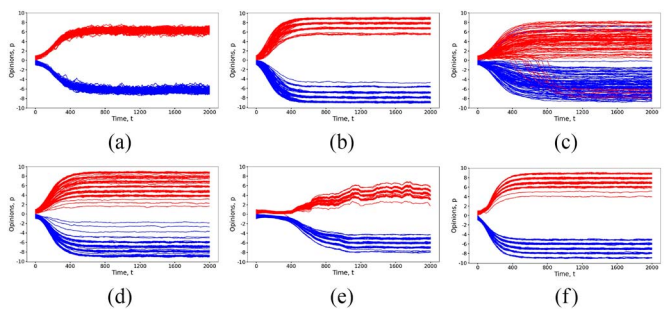

Fig. 5. Comparison of the evolution of opinions in six basic systems with half of the nodes being repulsive. The six subplots (a)–(f) respectively correspond to the temporal evolution over 2000 time steps under the intervention of recommendation algorithms based on content, users, relationships, popularity, and predetermined opinions.

in [1], α can serve as a representative of the controversial nature of event topics. The more controversial the event topic is, the more rapid the evolution of opinions becomes.

2) Impact of Repulsive Interaction on the Diversity of Opinion Distributions: In the microlevel model depicted in Fig. 3, which does not include any repulsive interaction rules, opinion evolution will tend to promote radicalization in one or two directions. However, if we enforce repulsive interaction rules for all nodes to replicate the extreme case of the backfire effect of misinformation, we find that the population will ultimately polarize, and this holds true for all types of recommendation algorithms. When repulsive and assimilative nodes are evenly distributed, we can observe the coexistence of opinions in both directions across all scenarios, as shown in Fig. 5. Notably, recommendation algorithms based on users and relationships are most conducive to supporting the survival of diverse and moderate opinions. These are the only two intervention scenarios that do not rely heavily on severe opinion homophily for evolution, and they also influence the rewiring outcomes between nodes. We further validate this intuition in Fig. 6. The precondition for the coexistence of diverse opinions in network structure manifests as individuals with opposing opinions maintaining connection relationships rather than causing isolated modules. Within each small community, assimilative nodes interact not only with nodes of the same camp, thereby partially converging towards opposing opinions and tending towards moderation. Conversely, repulsive nodes continue to approach one another when interacting with nodes of the same camp, dampening the trend towards radicalization. Consequently, we can witness a state of diversification.

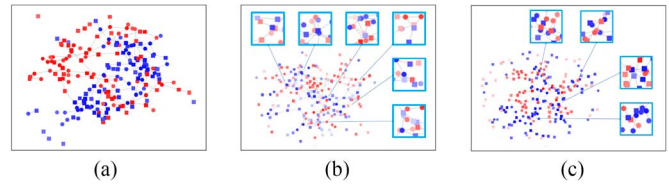

Fig. 6. Diverse opinions coexisting in networks: Subplots (a)–(c) respectively depict the final network under the intervention of content-based, users-based, and relationship-based recommendation algorithms. Each subplot consists of circular nodes representing assimilative nodes and square nodes representing repulsive nodes. The green box lists the cases where individuals with opposing opinions maintain connectional relationships.

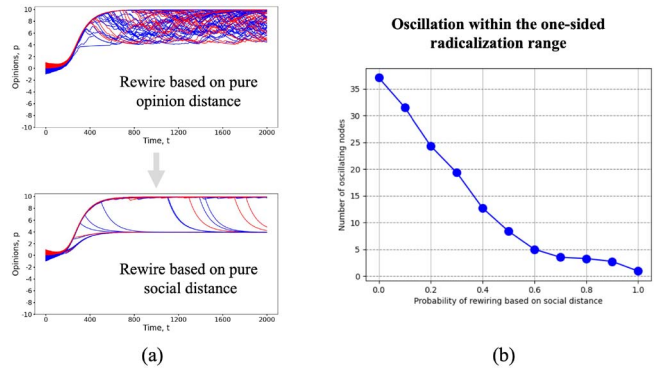

Fig. 7. Impact of rewiring rules. (a) In the evolutionary temporal comparison of 2000 time steps between rewiring rules based on pure opinion distance and pure social distance, it is observed that the final evolutionary outcomes exhibit a stable oscillation within a one-sided radicalization range, but there is a significant difference in the number of oscillating nodes within the radicalization range. (b) Impact of the probability of rewiring rules based on social distance on the number of oscillating nodes within the one-sided radicalization range, with the number of oscillating nodes averaged over 50 independent runs.

Summarizing Sections IV-C1 and IV-C2, we find that the conditions under which radical or moderate opinions emerge regarding highly controversial information, such as misinformation, are greatly influenced by external factors from the social environment and individual biases. In a highly controversial information space, even moderate opinions can exert significant social influence, leading to opinions becoming both radicalized and evolving rapidly. When we further consider the inherent backfire effect of misinformation, polarization becomes the normal state, unless the majority of nodes remain open-minded, willing to assimilate, and still allowed to interact with opposing opinions.

D. Impact of Network Evolution Parameters

1) Impact of Two Rewiring Rules on System Evolutionary Process: In our model, we design two types of rewiring rules based on opinion distance and social distance, and use parameters μ and ϕ to control their relative influence weights. By comparing the evolutionary processes of each system under

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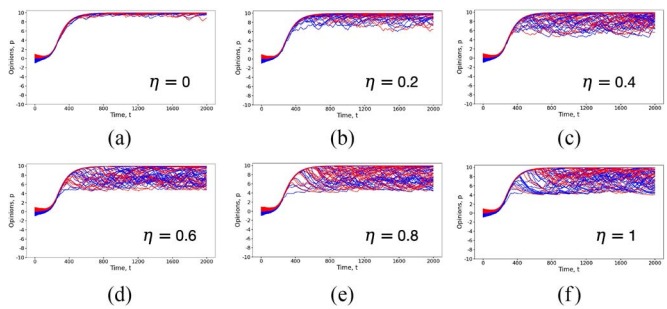

Fig. 8. Trend of system evolution process without recommendation mechanism as η varies. The six subgraphs (a)–(f) respectively correspond to the temporal evolution over 2000 time steps without algorithmic intervention for η values of 0, 0.2, 0.4, 0.6, 0.8, and 1.

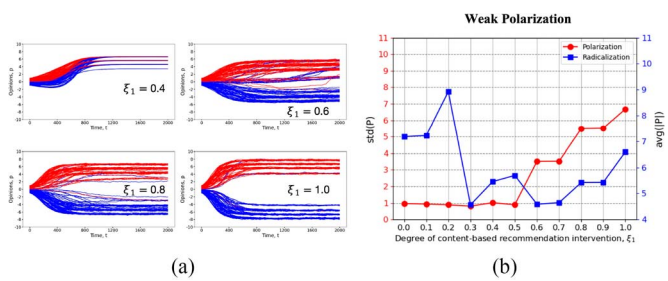

Fig. 9. Effects of the intervention of content-based recommendation in the hybrid recommendation scenario. (a) Trend of system evolution process under content-based recommendation intervention as ξ_1 varies. (b) Effect of the degree of content-based recommendation intervention in the hybrid recommendation scenario on polarization and radicalization levels. The red line represents changes in polarization levels, while the blue line represents changes in radicalization levels.

two configurations: rewire based on pure opinion distance (i.e., $\mu = 1,\ \phi = 0$) and rewire based on pure social distance (i.e., $\mu = 0,\ \phi = 1$), we observed no significant differences in the evolutionary processes of basic systems with recommendation algorithms. However, significant differences can be observed in the absence of recommendations. Furthermore, in Fig. 7, we can see that if nodes rewire more based on social distance, the lower the oscillation level within the stable radicalization range in the same direction they will exhibit. Actually, the influence of neighboring opinions remains continuous among interaction nodes in contact with the focal node, while the disturbance brought by exploration nodes is less likely to be sustained (due to the lower probability of rewiring). As a result, the adversarial conditions are weakened and most nodes tend to stabilize without further oscillation.

2) Impact of Rewiring Influence Strength on System Evolutionary Process: In this experiment, we set the rewiring probability to be equally dependent on opinion distance and social distance. We compare the microevolutionary processes of the six systems when the rewiring influence strength η equals 0 (i.e., equal probability of rewiring with each interactive node) and when η does not equal 0 (i.e., probability of rewiring depends on opinion distance and social distance, with η set to 1 in this case). The results revealed that there is no difference in the microevolutionary outcomes of the five systems with recommendation mechanisms under these two

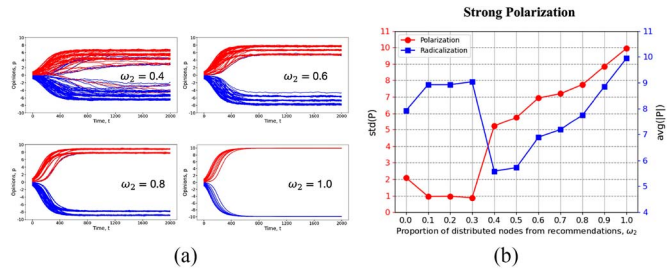

Fig. 10. Effects of intervention of recommendation distribution: (a) system evolution process under content-based recommendation intervention as ω_2 varies; (b) effect of the proportion of nodes from recommendation on polarization and radicalization levels. The red line represents changes in polarization levels, while the blue represents changes in radicalization levels.

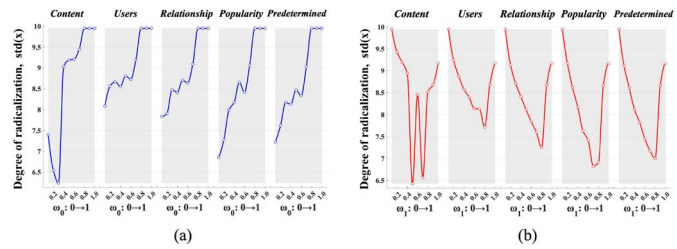

Fig. 11. (a) Variation of the degree of radicalization with ω_0 under five recommendation intervention system. (b) Variation of the degree of radicalization with ω_1 under five recommendation intervention system.

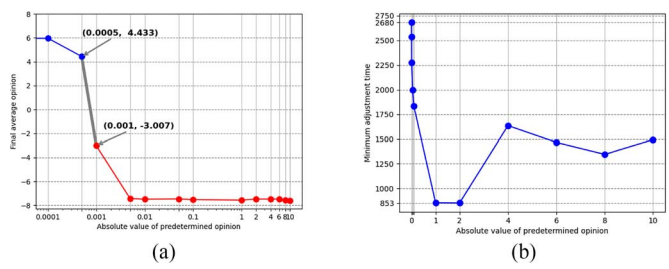

Fig. 12. Exploration of interventions based on predetermined information recommendations. (a) Effect of the absolute value of a predetermined opinion on the final population opinion, with the shaded bold gray line highlighting the threshold interval for opinion reversal (within the range of $[0.0005, 0.001]$). (b) Effect of the absolute value of a predetermined opinion on the minimum adjustment time (to reach a stable state of one-side radicalization).

conditions, while the systems without recommendations show different outcomes. With a step size of 0.2, we gradually vary η, within the range $[0, 1]$. We find that the final evolutionary state of the systems without any recommendation mechanisms gradually transitioning from a one-sided radicalized state to two independent modules, with a large number of nodes oscillating within stable radicalization ranges in the same direction (as shown in Fig. 8). According to (11), η serves as a decay exponent with an increase in η implying an exponential amplification of the influence of the rewiring rules based on opinion distance and social distance, as well as an exponential amplification of the adversarial influence brought by neighbors and exploration nodes. Therefore, oscillations within the system increase.

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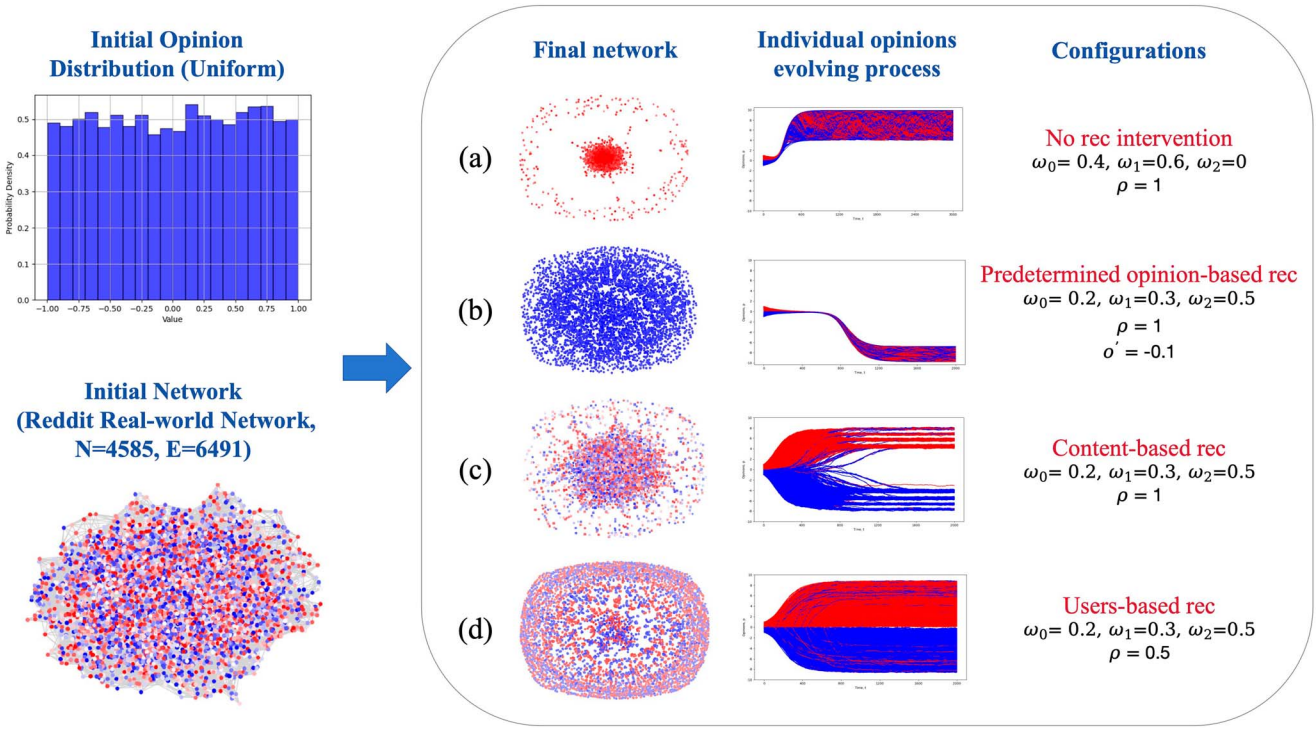

Fig. 13. Reproduction of four evolutional trends on a real-world network: The left side of the figure displays the initialization on the real-world Reddit network; the four subplots on the right (a)–(d) correspond to the temporal evolution over 2000 time steps under the different configurations, which reproduces the four evolutional trends and influence patterns we summarized in previous sections.

Summarizing Sections IV-D1 and IV-D2, we find that in the absence of algorithmic intervention, the network's evolution rules have a significant impact on the final outcome. However, once algorithms are introduced, they drastically alter the evolutionary trends, while the influence of the network's evolution rules becomes negligible. This underscores the necessity of a detailed discussion on the effects of algorithmic intervention.

E. Effects of Algorithmic Intervention

1) Effects of Each Basic Recommendation Algorithm in a Hybrid Recommendation Scenario: In this experiment, we implement a hybrid recommendation scenario where all five basic recommendation algorithms coexist. By gradually increasing the intervention proportion of a specific recommendation algorithm (while allocating the remaining proportion equally among the other algorithms), we are able to compare the influence of different types of algorithms on intervention degree. We find that content-based recommendations significantly counteract the trend of one-sided radicalization of opinions, while the intervention degree of other types of recommendations exerts little impact on the evolutionary outcomes. With the former increased in degree, the distribution of opinions shifts from being one-sided concentrated to divergent, eventually trending towards mild polarization (Fig. 9). This anomalous "weak polarization" effect may stem from constraints imposed by other sources of information. As long as interactions with exploration and neighboring nodes, which may bring about significant differences in opinions, continue to occur at a certain scale, the homophily and radicalization induced by content-based recommendation will not be too thorough.

2) Adjustments to the Algorithmic Distribution Mechanism: When adjusting the distribution mechanism by increasing the proportion of recommendation nodes in the interaction node set (ω_2), we find that the scenario of content-based recommendation tends to lead to "strong polarization" (as shown in Fig. 10). At this point, the utility of exploration and neighboring nodes becomes negligible. If we simply increase the proportion of nodes from exploration (ω_0), it is akin to random sampling, with an effect similar to that caused by popularity-based recommendation intervention, ultimately leading to radicalization in all scenarios. Conversely, if we solely increase the proportion of nodes from neighbors (ω_1), it becomes increasingly difficult to establish new connections, resulting in numerous isolated modules in the network and the emergence of many isolated opinion clusters, thereby making opinions more diverse and reducing the level of radicalization within a certain range (as shown in Fig. 11).

3) Promising Effect of Predetermined Opinions: In our numerous simulation experiments, it is found that the recommendation algorithm based on predetermined opinions possesses a dual power of reversal and invincibility, which appears to be the most promising intervention mechanism against misinformation. As shown in Fig. 12(a), enforcing a small bias (a rough threshold of -0.001) can alter the entire evolutionary trend (shifting the group's attitude towards misinformation from belief to disbelief), aligning with related findings on the nudge

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mechanism [49]. Moreover, we noted that a highly pronounced bias imposed (meaning a large absolute value of predetermined opinions) will constrain the minimum adjustment time to reach the final stable state, while a moderate bias acts better [as depicted in Fig. 12(b)].

Summarizing Sections IV-E1–IV-E3, we observe the disruptive power of algorithmic intervention on evolutionary trends—excessive reliance on content-based recommendations leads to weak polarization, while an overabundance of opinion distribution from recommendations triggers strong polarization. Allowing the information space to be dominated by exploration nodes without incorporating any recommendation reduces diversity, pushing the group towards radicalization. However, maintaining some visibility of neighbor information helps preserve opinion diversity. Lastly, the most subtle yet forceful nudge designs can easily shift the group's attitude toward misinformation from belief to disbelief.

F. Characterization of Real-World Network

By adjusting various parameters and directions of algorithmic interventions, we observed the temporal evolution of the opinions of the entire community in the Reddit real network under different configurations. Fig. 13 illustrates these results, showcasing trends and patterns consistent with those observed in synthetic networks in previous sections. Specifically, Fig. 13 reproduces scenarios such as oscillation within the one-sided radicalization range, one-sided radicalization, polarization, and the diversity of opinions. When the predetermined opinions were set to -0.01, we could still observe a reversal in the opinion evolution landscape, indicating the robustness of recommendation interventions based on predetermined information [Fig. 13(b)]. Introducing repulsive interaction rules in the real network also contributes to the diversity of opinions [Fig. 13(d)]. However, it should be noted that under the same configuration as the content-based recommendation intervention in Experiment IV-B1 [Fig. 3(b)], the polarization of opinions still occurs but is not as extreme [Fig. 13(c)]. Various interactions in complex real-world networks play a crucial role in this distinction.

V. Conclusion

We investigated the dissemination patterns of misinformation on social media and proposed a dynamic model that captures the coevolution process of user misinformation beliefs and social relationships under algorithmic intervention. In this model, misinformation tends to induce radicalization and provoke backfire effects. The establishment of social relationships is constrained by both opinion and social distances. Notably, algorithmic interventions act simultaneously on the information distribution process and the recommendation mechanism.

Through numerous simulations conducted at both micro and macrolevels, we unveiled intriguing behavioral patterns across these three elements with variable configurations. First, in our model, misinformation naturally evolves into radicalization, which addresses the skepticism mentioned in Section I: the natural evolutionary outcomes mediated by social processes without algorithms may themselves lead to radicalization. Solely observing the evolutionary process under a specific algorithmic intervention has limited significance and cannot determine its overall impact on misinformation diffusion. Additionally, evolutionary outcomes vary with changes in network evolution rules. In reality, these rules depend on individual biases, as individuals possess the decision-making right to establish or break connections, making them less susceptible to forced interventions by platforms. Our insights drawn from offline societies discuss the effects of opinion distance and social distance rules, align with previous work that specifically examines the influence of social–psychological mechanisms such as political bias [58], confirmation bias and peer pressure [59] on evolutionary outcomes. Our experiments are also supported by empirical data from Reddit, revealing consistent evolutionary trends and patterns across both synthetic and real networks. This confirms that our model is equally useful for quantitative and qualitative analyses of real systems, interpreting dynamics for real phenomena. The hypotheses we identified, which have critical implications for predicting misinformation diffusion and polarization, help bridge some gaps in understanding the emergence of polarized opinions in complex situations, especially considering the characteristics of misinformation. These further lays a solid foundation for empirical testing and strategic suggestions.

Another significant aspect of our work reveals the dual-edged impact of algorithmic intervention, providing pathways not only for understanding but also for solutions. We find that content-based recommendation is particularly flexible in altering evolutionary outcomes among various recommendation algorithms. Different levels of intervention can lead evolutionary results to a transition from one-sided radicalization to diversification and eventually to weak polarization. In other words, if the relevance between incoming information can be reduced, harmful effects like radicalization and polarization can be mitigated [60]. Furthermore, the significance of algorithmic distribution mechanisms is highlighted in our study. Any mechanism that isolates other information sources is prone to extreme outcomes, such as radicalization or strong polarization. These preliminary findings suggest that we have the capability to observe and control the radicalization and polarization of misinformation or promote diverse distributions. They also prompt us to consider how our proposed model could be implemented or promoted in real life. Platform designers can utilize our flexible framework to identify potential harmful scenarios, such as clarifying the consequences of introducing new features before deployment. Encouraging platforms to take action by adjusting different weight combinations in algorithmic distribution and recommendation can also aid in preventing misinformation dissemination and polarization. Finally, while we currently reveal promising prospects for nudge mechanisms operating at the level of opinion values, related mechanisms at a quantitative level are more practical in reality. This will be addressed in our future work.

Our work has several limitations. First, our model remains idealized, and future efforts should focus on collecting observational data to calibrate and validate the flexible model

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we have established. Second, the empirical dataset we currently use is not directly related to the themes of misinformation we are studying, although it provides substantial support for the microassumptions in our model and macropredictions related to polarization. This limitation arises from the challenge of collecting datasets that simultaneously encompass dynamic data on attitudes toward misinformation and networks of misinformation diffusion. Future research should consider integrating large-scale controlled experiments conducted online, and the emerging paradigm of agent-based testing (ABT) [61] has provided us with valuable insights. Third, our current simulation always observes a one-time, complete process under a specific set of parameters. However, these parameters could be adjusted more dynamically—for instance, by studying the impact of modifying algorithmic weights midway through the spread of misinformation. This approach could provide valuable insights for real-world emergency interventions to counter misinformation as it unfolds. Finally, our current work has explored the predictive capabilities and properties of our proposed innovative model, but we have yet to establish a robust indicator system for comparison with other similar models. Future plans will focus on assessing the impact of algorithmic interventions and using this as an evaluation metric to facilitate comparisons.

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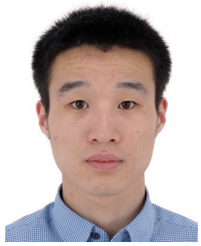

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